Introduction to Python Programming

Use python for...

- **Web Development**: Django, Pyramid, Bottle, Tornado, Flask, web2py
- **GUI Development**: tkInter, PyGObject, PyQt, PySide, Kivy, wxPython
- **Scientific and Numeric**: SciPy, Pandas, IPython
- **Software Development**: Buildbot, Trac, Roundup
- **System Administration**: Ansible, Salt, OpenStack, xonsh

Contents

- Introduction to programming
- Problem-solving (formulate - think - express)
- High-level languages
- Interpreters and compilers
- Example: The first python program

Printing Hello World in C++

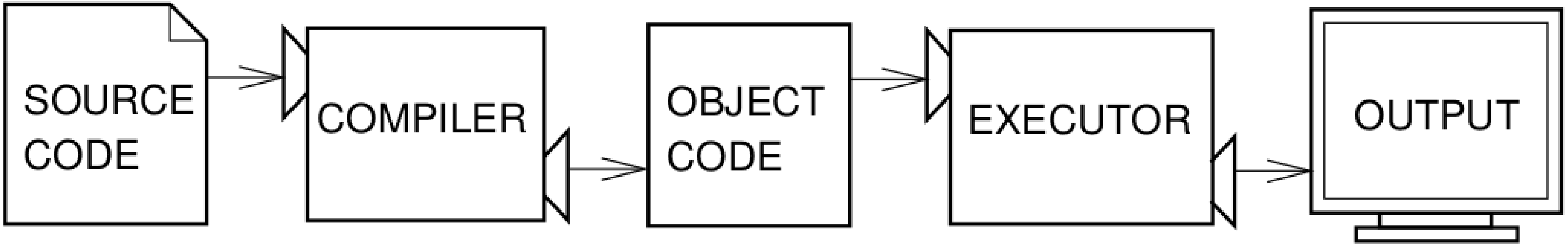
```
#include <iostream.h>

void main()
{
  cout << "Hello, world." << endl;
}
```

Printing Hello World in Python

- print "Hello, World!"

Interpreters & Compilers



Programming in Python

```
❶ passwordFile = open('SecretPasswordFile.txt')
❷ secretPassword = passwordFile.read()
❸ print('Enter your password.')
  typedPassword = input()
❹ if typedPassword == secretPassword:
❺    print('Access granted')
❻    if typedPassword == '12345':
❼       print('That password is one that an idiot puts on their luggage.')
  else:
❽    print('Access denied')
```

Math Operators – Interactive Shell

Operator	Operation	Example	Evaluates to...
**	Exponent	2 ** 3	8
%	Modulus/remainder	22 % 8	6
//	Integer division/floored quotient	22 // 8	2
/	Division	22 / 8	2.75
*	Multiplication	3 * 5	15
-	Subtraction	5 - 2	3
+	Addition	2 + 2	4

Common Data Types

Data type	Examples
Integers	-2, -1, 0, 1, 2, 3, 4, 5
Floating-point numbers	-1.25, -1.0, --0.5, 0.0, 0.5, 1.0, 1.25
Strings	'a', 'aa', 'aaa', 'Hello!', '11 cats'

```
>>> 'Hello world!
SyntaxError: EOL while scanning string literal
```

>>>‘NIE’ +’ ‘ +’Mysore’

‘NIE Mysore’

Storing values in variables

```
>>> 'shashank ' * 4
'shashank shashank shashank shashank '

>>> spam = 40
>>> spam
40

>>> eggs = 2
>>> spam + eggs
42

>>> spam + eggs + spam
82
```

```
>>> spam = spam + eggs
>>> spam
42

>>> spam = 'Hello'
>>> spam
'Hello'

>>> spam = 'Goodbye'
>>> spam
'Goodbye'
```

Variable Names

1) It can be only one word.
2) It can use only letters, numbers, and the underscore (_) character.
3) It can’t begin with a number.

Valid and invalid variable names

Valid variable names	Invalid variable names
balance	current-balance (hyphens are not allowed)
currentBalance	current balance (spaces are not allowed)
current_balance	4account (can't begin with a number)
_spam	42 (can't begin with a number)
SPAM	total_$um (special characters like $ are not allowed)
account4	'hello' (special characters like ' are not allowed)

Next Class

- Flow Control